

torch.nn.functional.avg_pool1d#

https://pytorch.org/docs/stable/generated/torch.nn.functional.avg_pool1d.html

Description:

Applies a 1D average pooling over an input signal composed of several input planes. See AvgPool1d for details and output shape. input – input tensor of shape $(\text{minibatch}, \text{in_channels}, iW)$, iW kernel_size – the size of the window. Can be a single number or a tuple $(kW,)$ stride – the stride of the window. Can be a single number or a tuple $(sW,)$. Default: kernel_size

Function Signature

```
torch.nn.functional.avg_pool1d(input, kernel_size,
stride=None, padding=0, ceil_mode=False,
count_include_pad=True) → Tensor#
```

Parameters

Code Examples

```
>>> # pool of square window of size=3, stride=2
>>> input = torch.tensor([[[1, 2, 3, 4, 5, 6, 7]]],
dtype=torch.float32)
>>> F.avg_pool1d(input, kernel_size=3, stride=2)
tensor([[[ 2.,  4.,  6.]])
```

Detailed Documentation

Documentation

Applies a 1D average pooling over an input signal composed of several input planes.

See AvgPool1d for details and output shape.

`input` – input tensor of shape $(\text{minibatch}, \text{in_channels}, iW)$, $(\text{minibatch}, \text{in_channels}, iW)$

`kernel_size` – the size of the window. Can be a single number or a tuple $(kW,)$

`stride` – the stride of the window. Can be a single number or a tuple $(sW,)$. Default: `kernel_size`

`padding` – implicit zero paddings on both sides of the input. Can be a single number or a tuple $(padW,)$. Should be at most half of effective kernel size, that is $((kernelSize - 1) * dilation + 1) / 2$. Default: 0

`ceil_mode` – when True, will use ceil instead of floor to compute the output shape. Default: False

`count_include_pad` – when True, will include the zero-padding in the averaging calculation. Default: True